2025

ELECTRIC 3 AXES MOTION SYSTEM NGS-360-3

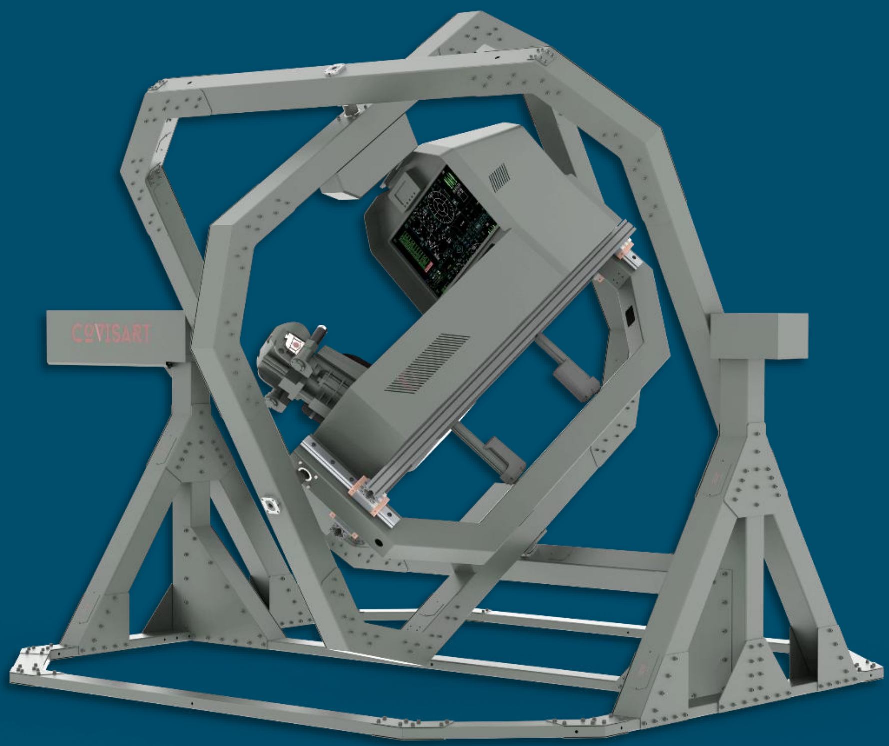
SPECIFICATIONS

NGS-360

COVISART

NGS-360-3 — Unlimited motion, unmatched precision

COVISART delivers next-generation electrically driven motion systems for modern simulation needs. The NGS-360-3 is built for mid-range payloads up to 800 kg, offering unlimited 3-axis rotation with 1 ms communication speed for ultra-precise response. Inspired by gyroscope technology, the system supports both 3-axis and multi-axis motion, ensuring realistic and dynamic simulation performance.



One system, endless vehicle possibilities.

The cockpit (gondola) is fully reconfigurable, allowing it to replicate any vehicle type required for training or research. This rapid adaptability minimizes both cost and time, enabling efficient execution of diverse simulation and testing programs.

The New Generation Simulator (NGS) extends beyond traditional motion training by addressing a broader range of challenges, including the study of pilot disorientation, human-factor dynamics, and other medical research applications related to spatial perception and motion effects.

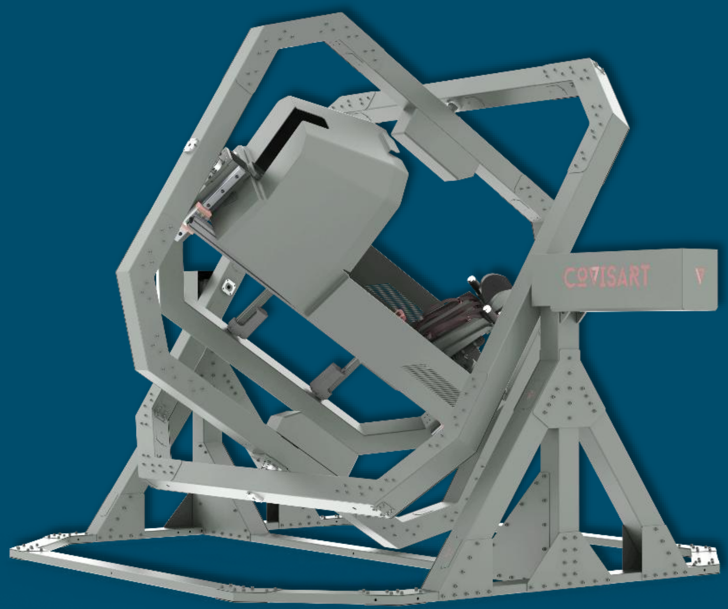
Typical Applications

- Driving Simulation – cars, trucks, buses, trains, trams, and off-road/rough terrain vehicles
- Flight Simulation – small aircraft and light helicopters
- Motion Rides – immersive entertainment and theme park applications

Advantages

- **Optimal Motion Control** – Powered by proprietary motion cueing software and full digital control for unmatched precision.
- **Best-in-Class Smoothness** – High-quality components, advanced friction compensation, ball screw spindles, oil lubrication, and low-friction seals ensure seamless operation.
- **Economic & Efficient Design** – High performance with low power consumption, achieved through optimized transmission ratios and integrated static compensation.
- **Optimized Motion Envelope** – Parallel-mounted Beckhoff servo motors with integrated brakes deliver responsive and reliable motion.
- **Safe Operation** – Built-in cushioning system, holding brakes, and advanced safety electronics/software guarantee operational security.
- **Fast Installation & Commissioning** – Compact subsystem architecture simplifies setup.
- **Fully Tested Delivery** – Units arrive completely assembled and tested for plug-and-play integration.

  Next Generation Motion Platform Manager – Intuitive control software enabling quick adjustments to motion, bumps, roll, and pitch with a single click.
- **Low-Maintenance Design** – Engineered for easy, low-interval servicing.



Scope of Supply

- Motion Base Assembly – Three electric servo motors, motion platform, base frame with integrated floor mounting plates
- System Electronics – Drive controllers and power units for reliable operation
- Motion Computer – Equipped with digital I/O and Ethernet communication interface
- Motion Control Software – Includes advanced motion cueing as well as ride-file play & replay capabilities
- User-Friendly Interface – Menu-driven design with remote access service and diagnostic tools
- Interconnecting Cabling – Complete wiring for plug-and-play integration
- Documentation Package – System Description, Interface Control Document, Operating and Maintenance Manuals

Options

- Access Ramp – Simplifies entry and exit to the simulator
- Extended Motion Capability – Upgradeable up to 5 axes for advanced simulation needs
- Alternative Host Interfacing – Options such as reflective memory for seamless integration
- User-Tunable Software – Customizable motion profiles and parameters
- Tailor-Made Motion Cueing – Fine-tuned cueing algorithms adapted to customer requirements
- Motion Ride File Creation – Development of custom ride scenarios for training or entertainment
- On-Site Services – Assembly, commissioning, fine tuning, maintenance, and operator training
- Payload Geometry Analysis – Envelope study tailored to the customer's specific payload setup
- Service & Maintenance Contracts – Long-term support solutions for sustained reliabilit

General Specifications

- Motor: Servo Motor – Integrated Brake
- Platform Shape: Square
- Ambient Temperatures: -10° to +45°C

Performance Specifications (Operational, non-simultaneous)

These figures are with the starting position 'neutral' and with the design payload.

Axis	Acceleration	Jerk	Velocity	Excursions (pos. / neg.)
Roll	+/- 220 deg/s²	+/- 600 deg/s²	+/- 220 deg/s	360 deg (unlimited)
Pitch	+/- 180 deg/s²	+/- 400 deg/s²	+/- 110 deg/s	360 deg (unlimited)
Yaw	+/- 110 deg/s²	+/- 200 deg/s²	+/- 90 deg/s	360 deg (unlimited)

Tablo 1

Dimensions and Weight

- Payload (load above platform): 500 kg
- Approx. Unit Weight: 1800 kg
- Height: 3100 mm
- Width (X): 3600 mm
- Width (Y): 3600 mm
- Area: 12 m²
- Footprint: 1800 x 1800 mm

Electrical Specifications

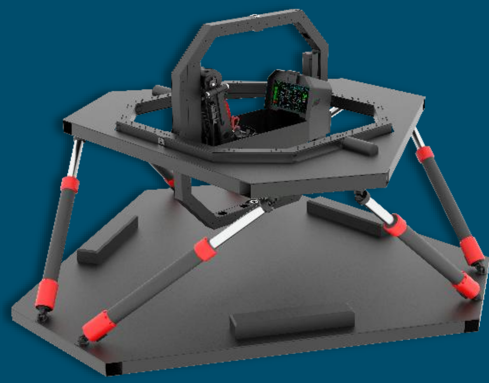
Axis	Power	Voltage - Ampere
Roll	4600 W	380v – 12A
Pitch	4700 W	380V – 12.4A
Yaw	4600 W	380v – 12A

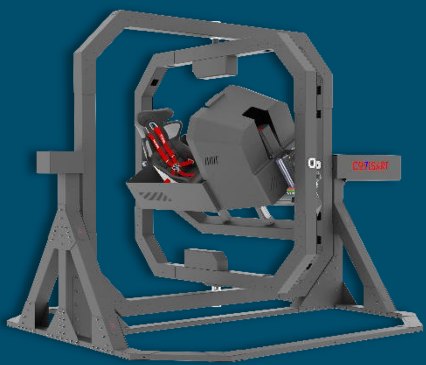
Interfaces

- Electric Power: 220/380 VAC, 50/60 Hz
- 4 Channel Data Transfer Module (30A – 380v)
- Host computer
- Ethernet connection
- Update rate up to 1 GHz

NGS OPTIONS

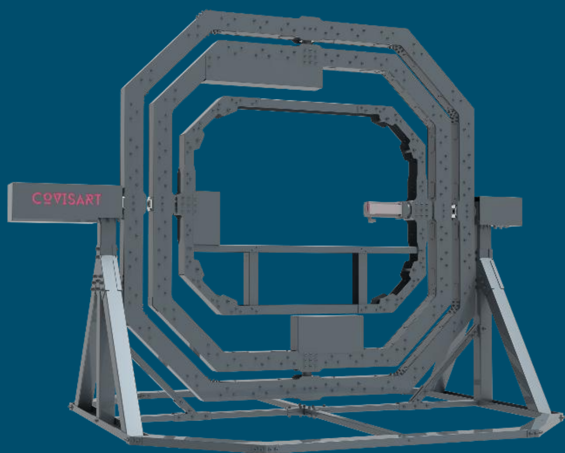
The NGS-360-6+2 is a high-end motion simulator that combines two unlimited rotation axes with a high-response 6-degrees-of-freedom (6DOF) system. Developed by the company Covisart, this simulator is designed for highly realistic training and entertainment applications.





The NGS-360-3+1 is a variant of the New Generation Simulator (NGS) motion platform. This system is designed to provide a highly realistic simulation experience by combining three axes of unlimited rotation with a single axis of translational movement. By combining the three rotational axes with the Z-axis (heave), simulator is capable of creating a powerful sensation of flight, ground vehicle movement, and other scenarios, offering a more immersive experience than systems with only rotational axes.

The NGS-360-3 is a foundational model in the NGS (New Generation Simulator) product line, developed by Covisart. As the name suggests, it is a 3-axis motion system with unlimited rotation capabilities. The NGS-360-3 is a more standard yet highly capable simulator compared to its more advanced variants. its unlimited rotation capabilities make it highly effective for simulating a broad range of vehicle dynamics where rotational cues are the most important part of the experience.



The NGS-360-2 is a more basic, entry-level version in the NGS simulator product line. It's designed to be a lower-cost option while still providing a highly immersive experience, particularly for applications where certain degrees of freedom are more critical than others. The NGS-360-2 is an example of how a motion simulator can be scaled down to a more accessible price point by focusing on the most critical axes of movement while retaining a key feature (unlimited rotation) to ensure a compelling and realistic experience.

Product Web Page

CONTACT:

Tel: +90 312 395 9639

Mail: info@covisart.com

Web: https://covisart.com.tr

Address 1: ANKARA IVEDIK OSB. TECHNOPARK ANKARA, TECHNOLOGY DEVELOPMENT ZONE

Address 2: IVEDIK OSB, 1466 Street, No:22. Yenimahalle/Ankara, Türkiye